连续与间断

连续

$$f(x)\text{在}x=a\text{的邻域内有定义}$$
$$\text{若}\lim_{x\to a}f(x)=f(a),\text{称}f(x)\text{在}x=a\text{连续}$$

1. $f(x)$在$x=a$连续 $\Leftrightarrow f(a-0)=f(a+0)=f(a)$
2. 初等函数在定义域内皆连续

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1. $f(x)$在(a,b)内点点连续
2. $f(a)=f(a+0)$(右连续), $f(b)=f(b-0)$(左连续)
   称$f(x)$在$[a,b]$上连续, 记$f(x)\in C[a,b]$

间断

$$\text{若}\lim_{x\to x_0}f(x)\neq f(x_0),x_0\text{为}f(x)\text{的间断点}$$

第一类间断点

$$f(x_0-0),f(x_0+0)\exists$$

$$\begin{cases}f(x_0-0)=f(x_0+0)(\neq f(x_0))-x_0\text{为可去间断点}\\ f(x_0-0)\neq f(x_0+0)-x_0\text{为跳跃间断点}\end{cases}$$

第二类间断点

$$f(x_0-0),f(x_0+0)\text{至少一个不存在}$$

$$\text{设函数}f(x)=\begin{cases}\frac{\sin ax+e^{2x}-1}{x},x<0\\ 4,x=0\\ \frac{b\arctan x+\ln(1-x)}{x},x>0\end{cases},\text{在}x=0\text{处连续, 求常数}a,b\text{的值.}$$

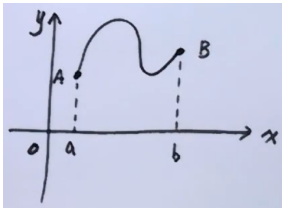
$$f(0-0)=\lim_{x\to 0^-}\frac{\sin ax}{x}+\lim_{x\to 0^-}\frac{e^{2x}-1}{x}=a+2$$
$$f(0)=4$$
$$f(0+0)=b\lim_{x\to 0^+}\frac{\arctan x}{x}+\lim_{x\to 0^+}\frac{\ln(1-x)}{x}=b-1$$
$$\because f(x)\text{在}x=0\text{连续},\therefore f(0-0)=f(0)=f(0+0)\Rightarrow a=2,b=5$$

$$\text{设}f(x)=\frac{x^2+x-2}{x^2-1}e^{\frac{1}{x}},\text{求函数}f(x)\text{的间断点并分类.}$$

$$x=-1,0,1\text{为间断点}$$
$$\lim_{x\to-1}f(x)=\infty\Rightarrow x=-1\text{为第二类间断点}$$
$$f(0-0)=\lim_{x\to0^-}\frac{x^2+x-2}{x^2-1}e^{\frac{1}{x}}=0$$
$$f(0+0)=+\infty$$
$$\Rightarrow x=0\text{为第二类间断点}$$
$$\lim_{x\to1}f(x)=\lim_{x\to1}\frac{(x-1)(x+2)}{(x-1)(x+1)}e^{\frac{1}{x}}=\lim_{x\to1}\frac{x+2}{x+1}e^{\frac{1}{x}}=\frac{3e}{2}\Rightarrow x=1\text{为可去间断点}$$

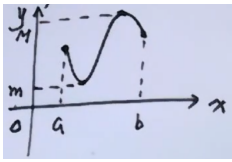
$$\text{设}f(x)=\frac{\ln|x|}{x^2-1},\text{求}f(x)\text{的间断点.}$$
$$x=-1,0,1\text{为间断点}$$
$$\lim_{x\to-1}f(x)=\lim_{x\to-1}\frac{1}{x-1}\frac{\ln(-x)}{x+1}=-\frac{1}{2}\lim_{x\to-1}\frac{\ln[1-(x+1)]}{x+1}=\frac{1}{2}$$
$$\Rightarrow x=-1\text{为可去间断点}$$
$$\lim_{x\to0}f(x)=+\infty\Rightarrow x=0\text{为第二类间断点}$$
$$\lim_{x\to1}f(x)=\lim_{x\to1}\frac{1}{x+1}\frac{\ln x}{x-1}=\lim_{x\to1}\frac{\ln[1+(x-1)]}{x-1}=\frac{1}{2}$$
$$\Rightarrow x=1\text{为可去间断点}$$

闭区间连续性质



$$f(x)\in C[a,b]$$

最值定理

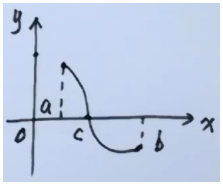


$$(m,M)\text{若}f(x)\in C[a,b]\Rightarrow f(x)\text{在}[a,b]\text{上取到最小值}m\text{和最大值}M$$

有界定理

$$\text{若}f(x)\in C[a,b]\Rightarrow\exists k>0,\text{使}|f(x)|\le k$$
$$\because f(x)\ge m,f(x)\le M,\therefore f(x)\text{在}[a,b]\text{上有界}$$

零点定理

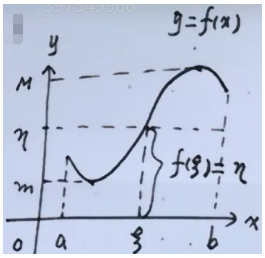


$$\text{若}f(x)\in C[a,b],\text{且}f(a)f(b)<0$$
$$\Rightarrow \exists c\in(a,b),\text{使}f(c)=0$$

$$\text{证明}x^5+4x-1=0\text{有且仅有一个正根.}$$

$$\text{证：令}f(x)=x^5+4x-1\in C[0,1]$$
$$f(0)=-1,f(1)=4$$
$$\because f(0)f(1)<0,\therefore \exists c\in(0,1),\text{使}f(c)=0$$
$$\because f'(x)=5x^4+4>0(x>0)$$
$$\therefore f(x)\text{在}[0,+\infty]\uparrow$$
$$\therefore \text{正根唯一}$$

介值定理



$$[m,M]-\text{值域}$$
$$\text{位于}[m,M]\text{之间的任一个值称为介值}$$
$$\forall\eta\in[m,M],\exists\xi\in[a,b]\text{使}f(\xi)=\eta$$
$$\text{位于}m\text{和}M\text{之间任何值}f(x)\text{皆可取到}$$

$$\text{设}f(x)\in C[a,b],\forall\eta\in[m,M],\exists\xi\in[a,b],\text{使}$$
$$f(\xi)=\eta$$

$$f(x)\in C[0,2],f(0)+2f(1)+3f(2)=6$$
$$\text{证：}\exists\xi\in[0,2],\text{使}f(\xi)=1$$

$$\text{证：}f(x)\in C[0,2]\Rightarrow\exists m,M$$
$$6m\le f(0)+2f(1)+3f(2)\le 6M$$
$$\Rightarrow m\le 1\le M$$
$$\exists\xi\in[0,2],\text{使}f(\xi)=1$$